

Embeddings, *P*-groups, and Σ^c_α Sentences

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Classification

In many branches of math, we would like to classify structures with respect to some notion of equivalence.

- Given classes of structures, can we determine which has a more difficult classification problem?
- If we can (*definably*) reduce a classification of K to one of K' , then the classification problem for K' should be 'more difficult.'
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Turing Computable Embeddings

Calvert, et al. considered only computable languages, and structures with universes subsets of ω , in order to formulate an effective analog.

Definition

A *Turing computable embedding* of (K, E) into (K', E') is an operator $\Phi = \phi_e$ such that

- for each $\mathcal{A} \in K$ there exists $\mathcal{B} \in K'$ such that $\Phi(\mathcal{A}) = \phi_e^{D(\mathcal{A})} = \chi_{D(\mathcal{B})}$, and
- if $\mathcal{A}, \mathcal{A}' \in K$, then $\mathcal{A}E\mathcal{A}' \leftrightarrow \Phi(\mathcal{A})E'\Phi(\mathcal{A}')$.

This induces a preordering, denoted by $(K, E) \leq_{tc} (K', E')$.

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Some Results

- $PF <_{tc} FLO <_{tc} FVS <_{tc} VS <_{tc} LO$.
- For all K , $K \leq_{tc} UG \equiv_{tc} LO$.
- $VS \equiv_{tc} ACF \equiv_{tc} ZS$
- If $(K, E) \leq_{tc} VS$, then E -classes of K are distinguished by Σ_2^C sentences.

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Computable Infinitary Sentences

The *computable infinitary formulas* are formulas of $\mathcal{L}_{\omega_1\omega}$ where all disjunction/conjunctions are computably enumerable.

Definition

- The Σ_0^c and Π_0^c formulas are the finitary quantifier-free formulas (we suppose they are always in normal form).
- For a computable ordinal $\alpha > 0$, $\phi(\bar{x})$ is a Σ_α^c formula if it is of the form

$$\bigvee_{i \in I} (\exists \bar{u}) \psi_i(\bar{u}, \bar{x}),$$

where each ψ_i is of lower complexity.

- The Π_α^c formulas are defined analogously.

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The Pull-back Theorem

A tool for obtaining necessary conditions for an embedding to exist is:

Theorem (Pull-back Theorem, Knight, Miller, Vanden Boom)

If $(K, E) \leq_{tc} (K', E')$ via Φ , then for any computable infinitary sentence ψ in the language of K' , we can (effectively) find a computable infinitary sentence ψ^ in the language of K such that*

- for all $\mathcal{A} \in K$, $\Phi(\mathcal{A}) \models \psi$ if and only if $\mathcal{A} \models \psi^$
- ψ and ψ^* have the same complexity.*

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P -Groups

- To each (reduced) Abelian p -group, we associate an *Ulm sequence*, an element of $(\omega + 1)^{<\omega_1}$.
- We can say $U_\alpha(\mathcal{G}) \geq k$ with a computable infinitary sentence.

Theorem (Ulm, 1933)

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Motivated by the importance of Σ_α^c sentences in many examples, we define the following equivalence relations.

Definition

- In any class K , satisfying our conventions, we define the equivalence relation

$$\mathcal{A} \sim_\alpha \mathcal{B} \leftrightarrow (\forall \phi \in \Sigma_\alpha^c)(\mathcal{A} \models \phi \leftrightarrow \mathcal{B} \models \phi).$$

- $\mathcal{A} \sim \mathcal{B}$ if and only if $(\forall \alpha < \omega_1^{CK})(\mathcal{A} \sim_\alpha \mathcal{B})$

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A Lemma

Let AB_α^p be the class of reduced Abelian p -groups of length α .

Lemma (V.)

For any class K , if $K \leq_{tc} AB_\omega^p$, then for any $\mathcal{A}, \mathcal{B} \in K$,

$$\mathcal{A} \cong \mathcal{B} \leftrightarrow \mathcal{A} \sim_2 \mathcal{B}.$$

Proof: Apply the Pull-back theorem to fact that members of AB_ω^p are distinguished by Σ_2^c sentences.

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Length ω Case

Theorem (V.)

For any class K , $(K, \sim_2) \leq_{tc} (AB_\omega^p, \sim_2)$.

Proof: We modify proofs of S. Quinn to manually build $\Phi = \phi_e$.
 First note that we can enumerate the Σ_2^c sentences:

$$\bigvee_{i \in A} (\exists \bar{u}_i) \bigwedge_{j \in B} (\forall \bar{v}_j) \psi_j(\bar{u}_i, \bar{v}_j).$$

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Proof, cont.

- We exploit the Σ_2 guessing strategy to build a copy of

$$\mathbb{Z}_p^{m_1} \oplus \mathbb{Z}_{p^2}^\omega \oplus \mathbb{Z}_{p^3}^{m_2} \oplus \mathbb{Z}_{p^4}^\omega \oplus \dots,$$

where all $m_i \in \{0, \omega\}$.

- So long as we think a given $\phi_n \in \Sigma_2^c$ is true in the input structure, make $m_{n+1} = \omega$ in the output group. If we find out we are wrong, reset it to 0 by trashing corresponding summand.

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Higher Lengths

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For any class K , $(K, \sim_n) \leq_{tc} (AB_{\omega \cdot m}^p, \sim_n)$ if and only if $n \leq 2m$.

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For any class K , $(K, \sim_\alpha) \leq_{tc} (AB_\beta^p, \sim_\alpha)$ if and only if $\beta = \omega \cdot \gamma$ and $\alpha < \gamma$.

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Motivation

- For *bounded* $\alpha < \omega_1^{CK}$, there is a certain length β such that $(K, \sim_\alpha) \leq_{tc} (AB_\beta^p, \sim_\alpha)$.
- Finding the appropriate β , and a $\Phi = \phi_e$ to witness the embedding seems very uniform.
- Can we ‘lace together’ all of these embeddings into a ‘master procedure,’ showing that $(K, \sim) \leq_{tc} AB^p$?

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Some Lemmas

Lemma (?)

There is a partial computable function $\cdot_{\mathcal{O}} : \mathcal{O} \times \mathcal{O} \rightarrow \mathcal{O}$ such that, for any $a, b \in \mathcal{O}$,

$$|a| \cdot |b| = |a \cdot_{\mathcal{O}} b|.$$

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There is a computable operator, $\Phi = \phi_e$, such that, given $(G_i)_{i \in I} = \{\langle i, \phi \rangle : i \in I, \phi \in D(G_i)\}$, we have

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A Uniform Result

Theorem (V.)

There is a computable function, $f : \mathcal{O} \rightarrow \omega \times \mathcal{O}$ such that, for $a \in \mathcal{O}$, $f(a) = \langle e, b \rangle$, where ϕ_e is the operator witnessing $(K, \sim_{|a|_{\mathcal{O}}}) \leq_{tc} AB_{|b|_{\mathcal{O}}}^p$.

Proof Sketch: We have a function on ordinals which, when given $|a|_{\mathcal{O}}$, outputs the length of groups we need to code.

$$g(\omega \cdot \beta + n) = g(\beta, n) = \begin{cases} \omega^2 \cdot \beta + \omega \cdot \frac{n}{2} & \text{if } n \text{ is even} \\ \omega^2 \cdot \beta + \omega \cdot \frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases}$$

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Barwise-Kreisel Compactness

Theorem (Barwise, Kreisel)

Let Γ be a Π_1^1 set of computable infinitary sentences (in a fixed computable language). If every Δ_1^1 subset of Γ is consistent, then Γ is consistent.

- Let Γ_α be the collection of sentences saying an operator, Φ , witnesses $(K, \sim_\alpha) \leq_{tc} AB_{g(\alpha)}^p$.
- $\forall \mathcal{A}, j[(\mathcal{A} \in K \wedge \mathcal{A} \models \psi_j) \leftrightarrow (U_{g(\alpha, j)}(\Phi(\mathcal{A})) = \omega)]$

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Unfortunately, we can only do this for certain X .

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